

uses embedded vision to track movements of Xbox 360 users, rather than requiring handheld controllers. Over the next few years, there will be a rapid proliferation of embedded vision technology into many kinds of embedded systems. Some of the key sectors worth looking up are retail, entertainment and medical.

Adding vision to electronics products

Brian Dipert, senior analyst, Berkeley Design Technology, noted, "The members of the Embedded Vision Alliance certainly believe that "vision technologies can add value capabilities to electronic products".

The following are some interesting case studies:

1. A 'smart' surveillance system that analyses the area in front of its cameras and captures footage only when particular situations arise, such as motion is detected (perhaps only that of human beings). Such a system could also automatically alert its owner to the potential problem situation via e-mail, text message, etc. One example could be a system that constantly monitors a swimming pool and sounds an alarm if it detects people struggling and in the danger of drowning.

2. Automotive driver assistance systems that, for example, alert the driver to an impending collision with objects ahead and even automatically slam on the brakes or take other evasive action. The system could also inform the driver about important warning or other signs on the roadway ahead or on either side of the vehicle. It could even signal when the driver is about to make an unintended lane transition due to lack of attention and take over the steering wheel to correct the lane change.

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3. Facial recognition systems that automatically 'unlock' a smartphone and load particular account settings when they recognise the person in front of the camera lens.

4. Gesture interfaces like that in the Microsoft Kinect for the Xbox 360, which give a simpler and more intuitive interactive experience for users of a variety of equipment, not just gaming consoles.

5. Medical instrumentation that automatically senses and logs respiration rate, heart rate and other vital parameters, and also analyses X-ray and other images for anomalies, helping physicians in diagnosis of the disorders.

6. Manufacturing-line automated control and defect analysis equipment, formerly implemented using dedicated high-end workstation computers and operating systems but now possible in much simpler, more rugged, compact and inexpensive forms.

Smart surveillance systems are already on their way. There are certain wireless-based security cameras that are designed for remotely monitoring the house and alert the user to potential emergencies via e-mail, text message and so on. Gaming devices such as the Wii and Microsoft Xbox Kinect also use embedded vision. A few elec-

tronic devices incorporate facial recognition systems to identify the persons using them. There are certain medical devices as well that can see and sense!

Embedded vision finds applications in diverse fields including automotive safety, machine vision, military and aerospace, and so on.

Tom Hackenberg, Semiconductors, IMS Research, forecasts the total market for embedded vision applications, including automotive safety and driver assistance, machine vision, security, military and aerospace, gesture recognition, digital signage and augmented reality usage, to grow from less than 15 million units in 2011 to over 60 million units in 2016.

In a recently published report 'The Indian Market for CCTV and Video Surveillance Equipment,' IMS Research, now acquired by IHS Inc., forecasts that city surveillance will be the fastest growing vertical market in India over the next five years. To counter terrorism, surveillance is now a key requirement.

Jon Cropley, principal analyst, IMS Research, forecasts video surveillance revenues for Indian airports to almost quadruple between 2011 and 2016. This isn't surprising at all. With growth rates averaging over 20 per cent during the next five years, India should continue to offer suppliers excellent growth opportunities. Here cloud-based video surveillance could also come into play!

IMS Research estimates that the global market for cloud-based video surveillance or video surveillance as a service was worth over \$500 million in 2011. The market could well exceed \$1 billion by 2014. That would be worth an eyeball! ●

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